Exam 2 Topics and Links

Lecture 15 Neural Networks (NN)

- History of Neural Network
- Multilayer Neural Network
- Forward propagation
- Backward propagation
- Optimization Algorithms
- Perceptron
- Multilayer perceptron
- Activation Functions
- Universal Approximation
- Loss Function
- Gradient Descent
- Vanishing Gradients/Exploding Gradients

- The Essential Main Ideas of Neural Networks
- Neural Networks Pt. 2: Backpropagation Main Ideas
- Neural Networks Pt. 3: ReLU In Action!!!
- Neural Networks Pt. 4: Multiple Inputs and Outputs
- Neural Networks Part 5: ArgMax and SoftMax
- Neural Networks Part 6: Cross Entropy
- Neural Networks Part 7: Cross Entropy Derivatives and Backpropagation
- But what is a neural network? | Deep learning chapter 1
- Neural Networks Explained: Step-by-Step Walkthrough
- Intro to Neural Networks : Data Science Concepts
- 12a: Neural Nets
- Neural Networks - Lecture 5 - CS50's Introduction to Artificial Intelligence with Python 2020
- Lecture 1.1 — Why do we need machine learning — [Deep Learning | Geoffrey Hinton | UofT]
- Lecture 10 - Introduction to Neural Networks | Stanford CS229: Machine Learning (Autumn 2018)
- Lecture 1.1 — Why do we need machine learning [Neural Networks for Machine Learning]

- Why Neural Networks can learn (almost) anything
- 10.4: Neural Networks: Multilayer Perceptron Part 1 - The Nature of Code
- What are MLPs (Multilayer Perceptrons)?
- 10.5: Neural Networks: Multilayer Perceptron Part 2 - The Nature of Code
- Neural Networks Demystified [Part 2: Forward Propagation]
- Forward Propagation in Neural Networks | Deep Learning
- Backpropagation, intuitively | Deep Learning Chapter 3
- Forward Propagation in a Deep Network (C1W4L02)
- Backpropagation calculus | Deep Learning Chapter 4
- Backpropagation : Data Science Concepts
- Neural Networks Pt. 2: Backpropagation Main Ideas
- The Most Important Algorithm in Machine Learning
- 10.4: Neural Networks: Multilayer Perceptron Part 1 - The Nature of Code
- What are MLPs (Multilayer Perceptrons)?
- 10.5: Neural Networks: Multilayer Perceptron Part 2 - The Nature of Code
- Neural Networks Explained: Building a Multilayer Perceptron!
- Activation Functions - EXPLAINED!
- Activation Functions In Neural Networks Explained | Deep Learning Tutorial
- Why Do We Need Activation Functions in Neural Networks?
- Activation Functions (C1W3L06)
- Activation Functions in a Neural Network explained
- The Universal Approximation Theorem for neural networks
- Visual Proof: How Neural Networks Can Solve Anything | Universal Approximation Theorem
- What is a Loss Function? Understanding How AI Models Learn
- Loss in a Neural Network explained
- Loss Functions - EXPLAINED!
- Loss functions in Neural Networks - EXPLAINED!
- Which Loss Function, Optimizer and LR to Choose for Neural Networks
- https://www.youtube.com/watch?v=IHZwWFHWa-w
- Gradient Descent Explained
- Gradient Descent in 3 minutes

- Gradient Descent, Step-by-Step
- Machine Learning Crash Course: Gradient Descent
- Gradient Descent vs Evolution | How Neural Networks Learn
- Vanishing & Exploding Gradient explained | A problem resulting from backpropagation
- Vanishing and exploding gradients | Deep Learning Tutorial 35 (Tensorflow, Keras & Python)
- Vanishing/Exploding Gradients (C2W1L10)
- The Fundamental Problem with Neural Networks - Vanishing Gradients

Lecture 16 Convolutional Neural Networks (CNN)

- Architecture Overview
- Layers Used to Build CNN
- Convolutional Layer
- Convolutional Layer- filter size
- Convolutional Layer - Stride
- Convolutional Layer - Padding
- Activation Layer
- Normalization Layer
- Pooling
- Pooling Layer
- Fully Connected Layer
- Several Common Architectures
- Applications

- What are Convolutional Neural Networks (CNNs)?
- But what is a convolution?
- Neural Networks Part 8: Image Classification with Convolutional Neural Networks (CNNs)
- CNN: Convolutional Neural Networks Explained - Computerphile
- Convolutional Neural Networks (CNNs) - Explained
- Convolutional Neural Networks Explained (CNN Visualized)
- Convolutional Neural Networks (CNNs) explained
- Why do Convolutional Neural Networks work so well?
- MIT 6.S191 (2025): Convolutional Neural Networks
- Lecture 5 | Convolutional Neural Networks

- Neural Networks - Lecture 5 - CS50's Introduction to Artificial Intelligence with Python 2020
- CS231n Winter 2016: Lecture 7: Convolutional Neural Networks
- Lecture 7: Convolutional Networks
- Filter Count - Convolutional Neural Networks
- 2D Convolution Explained: Fundamental Operation in Computer Vision
- ml5.js: What is a Convolutional Neural Network Part 1 - Filters
- Stride in Convolutional Neural Network (CNN)
- Stride - Convolution in Neural Networks
- Padding in Convolutional Neural Network
- Tutorial 22- Padding in Convolutional Neural Network
- Convolution Padding - Neural Networks
- C4W1L04 Padding
- Zero Padding in Convolutional Neural Networks explained
- Fully Connected Layer in CNN
- Neural Networks Pt. 3: ReLU In Action!!!
- Visualizing Convolutional Neural Networks | Layer by Layer
- What is Layer Normalization? | Deep Learning Fundamentals
- Batch Normalization ("batch norm") explained
- Batch normalization | What it is and how to implement it
- Normalizing Activations in a Network (C2W3L04)
- What is pooling? | CNN's #3
- Max Pooling in Convolutional Neural Networks explained
- Max Pooling in Convolutional Neural Network
- Tutorial 24- Max Pooling Layer In CNN
- Convolutional Neural Networks from Scratch | In Depth
- Convolutional Neural Network from Scratch | Mathematics & Python Code
- Backpropagation in Convolutional Neural Networks (CNNs)

Lecture 17 Long Short-Term Memory (LSTM)

- Recurrent Neural Network
- Long Short-Term Memory (LSTM)
- Cell State
- Forget Gate
- Input Gate
- Output Gate

- LSTM Training
- LSTM Applications
- Bidirectional RNN

- Long Short-Term Memory (LSTM), Clearly Explained
- What is LSTM (Long Short Term Memory)?
- Long Short-Term Memory for NLP (NLP Zero to Hero - Part 5)
- Deep Learning: Long Short-Term Memory Networks (LSTMs)
- Recurrent Neural Networks (RNNs), Clearly Explained!!!
- Long Short-Term Memory with PyTorch + Lightning
- Illustrated Guide to LSTM's and GRU's: A step by step explanation
- LSTM Networks: Explained Step by Step!
- Lecture 7.5 — Long term Short term memory — [Deep Learning | Geoffrey Hinton | UofT]
- Illustrated Guide to Recurrent Neural Networks: Understanding the Intuition
- A friendly introduction to Recurrent Neural Networks
- The Power of Recurrent Neural Networks (RNN)
- The Idea That Made RNNs Work
- Recurrent Neural Networks : Data Science Concepts
- Bidirectional RNN | Deep Learning Tutorial 38 (Tensorflow, Keras & Python)
- Bidirectional RNN | Deep Learning | In-depth Explanation & Equations
- Illustrated Guide to Recurrent Neural Networks: Understanding the Intuition
- Bidirectional RNN Indepth Intuition- Deep Learning Tutorial

Lecture 18 Self-Supervised Learning (SSL)

- Supervised vs. unsupervised vs. self-supervised
- Self-Supervised Learning
- Phase-1: Pretraining
- Phase-II: Fine-Tune
- Autoencoder
- predict rotations
- predict relative position
- Contrastive Learning
- Contrastive Representation Learning

- SimCLR
- BYOL
- Multimodal Contrastive Learning

- What Is Self-Supervised Learning and Why Care?
- How AI Taught Itself to See [DINOv3]
- What is Self Supervised Learning?
- Contrastive Learning with SimCLR | Deep Learning Animated
- What is Semi-Supervised Learning?
- Yann LeCun: Self-Supervised Learning Explained | Lex Fridman Podcast Clips
- Introducing DINOv3: Self-supervised learning for vision at unprecedented scale
- What is Self-Supervised Learning? [Quick Introduction]
- Stanford CS231N | Spring 2025 | Lecture 12: Self-Supervised Learning
- Stanford CS229 Machine Learning I Self-supervised learning I 2022 I Lecture 16
- Yann LeCun | Self-Supervised Learning, JEPA, World Models, and the future of AI
- Self-supervised vision
- Supervised vs. Unsupervised Learning
- Self-Supervised Pretraining Improves Self-Supervised Pretraining
- DINO: Emerging Properties in Self-Supervised Vision Transformers (paper illustrated)
- What are Autoencoders?
- Autoencoders | Deep Learning Animated
- Masked Autoencoders Are Scalable Vision Learners – Paper explained and animated!
- Masked Autoencoders (MAE) Paper Explained
- Masked Autoencoders are Scalable Vision Learners Paper Explained in 5 Minutes!
- Contrastive Learning with SimCLR | Deep Learning Animated
- Contrastive Learning - 5 Minutes with Cyrill
- Supervised Contrastive Learning
- Contrastive Learning in PyTorch - Part 1: Introduction
- Can Contrastive Learning Work? - SimCLR Explained
- Contrastive Loss : Data Science Basics

- BYOL: Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning (Paper Explained)
- Fixing SimCLRs Main Problem - BYOL Paper Explained
- Data Learning: Towards Better Understanding of Contrastive Learning
- Supervised Contrastive Learning
- Yann LeCun: Self-Supervised Learning Explained | Lex Fridman Podcast Clips
- Contrastive Learning with SimCLR | Deep Learning Animated
- How DINO learns to see the world - Paper Explained
- Denoising Autoencoders | Deep Learning Animated
- Autoencoders | Deep Learning Animated
- Yann LeCun - Self-Supervised Learning: The Dark Matter of Intelligence (FAIR Blog Post Explained)
- Supervised Contrastive Learning

Lecture 19 Transformer

- The Vanilla Transformer
- Input/Output Embedding
- Positional Encoding
- Self-Attention
- Multi-Head Attention
- Feed Forward Layer
- Add & Norm Layer
- Creating Masks

- What are Transformers (Machine Learning Model)?
- Transformers, explained: Understand the model behind GPT, BERT, and T5
- Transformers, the tech behind LLMs | Deep Learning Chapter 5
- Transformer Neural Networks, ChatGPT's foundation, Clearly Explained!!!
- Transformer Explained
- Lec 08. Architectures: Transformers
- 7: Deep Learning for Natural Language – Transformers
- Transformers architecture mastery | Full 7 hour compilation
- Transformer Neural Networks Derived from Scratch
- What Transformers Actually See (Tokens vs Embeddings)

- Transformer Explainer- Learn About Transformer With Visualization
- Attention in transformers, step-by-step | Deep Learning Chapter 6
- Transformers and Self-Attention (DL 19)
- Self-attention in deep learning (transformers) - Part 1
- Attention mechanism: Overview
- Self Attention in Transformer Neural Networks (with Code!)
- A Dive Into Multihead Attention, Self-Attention and Cross-Attention
- Multi Head Attention in Transformer Neural Networks with Code!
- Visual Guide to Transformer Neural Networks - (Episode 1) Position Embeddings
- Visual Guide to Transformer Neural Networks - (Episode 2) Multi-Head & Self-Attention
- Visual Guide to Transformer Neural Networks - (Episode 3) Decoder's Masked Attention
- How Attention Got So Efficient [GQA/MLA/DSA]
- Query, Key and Value Matrix for Attention Mechanisms in Large Language Models
- I Visualised Attention in Transformers
- Why the name Query, Key and Value? Self-Attention in Transformers | Part 4
- Self-Attention in Transformers | Q K V Explained for Beginners
- Simplest explanation of Layer Normalization in Transformers
- E08 Normalization (Batch, Layer, RMS) | Transformer Series (with Google Engineer)
- What is Layer Normalization? | Deep Learning Fundamentals
- E07 Feed Forward Network | Transformer Series (with Google Engineer)
- Why Transformers Use Feedforward Layers | Explained Visually
- Neural Networks Pt. 3: ReLU In Action!!!
- A Review of 10 Most Popular Activation Functions in Neural Networks
- The math behind Attention: Keys, Queries, and Values matrices
- How might LLMs store facts | Deep Learning Chapter 7
- What are Transformer Models and how do they work?
- The Attention Mechanism in Large Language Models
- Keys, Queries, and Values: The celestial mechanics of attention
- TFIDF : Data Science Concepts

- What Are Word Embeddings?
- Attention for Neural Networks, Clearly Explained!!!
- Attention is all you need (Transformer) - Model explanation (including math), Inference and Training
- The Transformer Explained: A Complete Layer-by-Layer Visual Breakdown

Lecture 20 Outlier Detection

- Anomaly/Outlier Detection:
- Distinction Between Noise and Anomalies
- Supervised, semi-supervised, and unsupervised outlier detection
- Types of Outliers
    - Point Anomalies
    - Contextual Anomalies
    - Collective Anomalies
- Anomaly Detection Techniques
    - Statistical Methods
        - Z-Score/Standard Score
        - IQR
    - Distance-Based Methods
        - K-Nearest Neighbors (KNN)
        - Local Outlier Factor (LOF)
        - DBSCAN (Density-Based Spatial Clustering)
    - Cluster-Based Methods
        - K-Means
        - Gaussian Mixture Models
    - Machine Learning-Based Methods
        - One-Class SVM
        - PCA
        - Autoencoders
        - GAN
    - Time-Series Anomaly Detection
        - Moving Average
        - LSTM Autoencoder / RNNs
- Reconstruction Error
- Data reconstruction

- Complete Anomaly Detection Tutorials Machine Learning And Its Types With Implementation | Krish Naik
- Isolation Forests: Identify Outliers in Data
- What is the Local Outlier Factor
- Anomaly detection 101
- Anomaly Detection : Time Series Talk
- Mastering ANOMALY DETECTION in Machine Learning
- Introduction to Anomaly Detection for Engineers
- Outlier & Anomaly Detection using Isolation Forest | What are Anomalies? | What is Isolation Forest?
- Outlier Detection Methods: Supervised, Semi Supervised, Unsupervised, Proximity Based, Clustering Ba
- Isolation Forests: Identify Outliers in Data
- Anomaly Detection: Algorithms, Explanations, Applications
- What is the Local Outlier Factor
- Anomaly detection (Local Outlier Factor)
- Local Outlier Factor- Everything you need to know! | Outlier Detection| Machine Learning Algorithms
- #147 - Understanding One-Class SVM: A Deep Dive into Anomaly Detection
- One-Class SVM for Outlier Detection
- What Role Does Reconstruction Error Play In VAE Anomaly Detection?
- 180 - LSTM Autoencoder for anomaly detection
- Anomalies, Breaks, and Outliers Detection in Time Series
- Most comprehensive demo of outlier detection algorithms
- Introduction to Outlier Detection Methods - Wolfram Livecoding Session
- Outliers in Data Analysis... and how to deal with them!
- Anomaly Detection : Time Series Talk
- Robust Anomaly Detection + Seasonal-Trend Decomposition : Time Series Talk
- Time Series Anomaly Detection Techniques for Predictive Maintenance
- Anomaly Detection For Time Series Data in Python